Albert Wang

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Education

Stanford University — Sept. 2026 – March 2028
M.S. Electrical Engineering, Integrated Circuits — *Stanford, CA*

Columbia University & Bates College | 3-2 Dual Degree Program — Aug. 2021 – May 2026
B.S. Electrical Engineering (Columbia) | GPA: 3.72/4.0 | Dean's List 2024–2026 — *New York, NY*
B.A. Engineering, Minor in Philosophy (Bates) | GPA: 3.97/4.0 | Dean's List 2021–2024 — *Lewiston, ME*

Technical Skills

IC Design: CMOS transistor sizing, op-amp/filter design, small-signal analysis, biasing, stability; Cadence Virtuoso (schematic/layout/DRC/LVS/extraction), LTspice, Synopsys Design Compiler, PrimeTime
HDL & Digital: Verilog, VHDL, SystemVerilog; RTL/gate-level simulation, timing analysis, FSM design, synthesis
Lab & PCB: Oscilloscope, function generator, DMM, soldering; Altium, Autodesk Fusion 360, Autodesk Revit
Programming: Python, C/C++, MATLAB, Java; signal processing, data analysis
Languages: English (native), Chinese (fluent)

Projects

Two-Stage CMOS Op-Amp with Miller Compensation — Fall 2025, *New York, NY*
- Designed two-stage OTA with Miller compensation: gain = 4 V/V, 76 dB open-loop, 47° phase margin, 203 kHz CLBW.
- Optimized transistor sizing for 700 μA budget; <0.1% DC accuracy, 5.4 μs settling time, <6.5% overshoot.
- Performed AC/DC/transient analysis in **Cadence Spectre**; verified stability across PVT corners.

Full-Custom 65nm VLSI Microprocessor — Fall 2025, *New York, NY*
- Designed complete 8-bit microprocessor schematic-to-layout in 65nm CMOS using **Cadence Virtuoso**; datapath includes ALU, shifter, 8×8 SRAM, decoder, and PLA control unit.
- Achieved 100 MHz with 88.8% timing margin, 4,100 μm^2 area; completed full DRC/LVS verification; documented parasitic characterization (2–3× delay increase) in 18-page technical report.
- Post-layout power optimization: 876 μW total consumption; analyzed process, voltage, and temperature (PVT) sensitivity.

Class D AM Transmitter — Spring 2026, *New York, NY*
- Designed RF input buffer, complementary Class D switching pair, and series-LC output tank for a 1 MHz AM transmitter on a single 6 V supply; selected tank $Q = 10$ from the **FCC Part 15** out-of-band emission limits.
- Achieved 84 mW DC dissipation, 71% modulation depth, 3.4% THD, and >28 dB carrier harmonic suppression (8 dB FCC margin); verified in **LTspice**.

CLACS Smart Pen – Senior Design Project — Spring 2026, *New York, NY*
- Designed LiPo battery/USB-C charging subsystem and revised STM32WB55 MCU schematic in **Altium**; co-developed double-sided PCB with ∼84% area reduction.
- Hand-soldered SMD boards (IPC-J-STD-001); wrote I2C firmware that isolated solder-bridge and USB-C routing faults.
- Wrote full **C** middleware for live IMU streaming over USB-CDC/BLE 5.4; adaptive stroke detection improved segmentation ∼3× over baseline.

64-Tap FIR Filter with Clock Domain Crossing — Fall 2025, *New York, NY*
- Designed 64-tap 16-bit FIR filter in **Verilog** with asynchronous FIFO, MAC with saturation, FSM controller.
- Synthesized using **Synopsys Design Compiler** (IBM 130nm) achieving 100 MHz timing closure, 46,551 μm^2 area; validated via gate-level simulation with SDF back-annotation.
- Power analysis via **PrimeTime PX**: 3.384 mW total, 0.034 mW/MHz efficiency, 100% VCD annotation.

Experience

Hardware Security Intern (PUF/HSM) — July 2025 – September 2025
eMemory Technology Inc. — *Hsinchu, Taiwan*
- Co-authored internal API specification for PUF-backed HSM framework covering PKCS#11 partitions, HA/backup, and operations; document adopted as reference for **FIPS 140-3** certification testing.

Collegiate MEP Intern — May 2025 – July 2025
PBK Architects (LEAF Engineers) — *Houston, TX*
- Developed NEC-compliant panel schedules and circuiting in **Autodesk Revit** for large-scale institutional building electrical systems; performed load calculations, feeder/breaker sizing, and phase balancing.

Research

NSF-REU Computational Physics Researcher — May 2024 – October 2024
Clarkson University — *Potsdam, NY*
- Developed projection-based learning algorithm in **MATLAB** achieving 100× speedup for photonic crystal simulations.
- Co-authored peer-reviewed paper at SPIE Photonics West 2025 (DOI: 10.1117/12.3028208).